CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 16-2026: A 14-Year-Old Girl with Hypertension

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CME

PRESENTATION OF CASE

Dr. Nora AlFakhri (Pediatrics): A 14-year-old girl was admitted to this hospital because of hypertension.

The patient had been in her usual state of health until 10 days before the current admission, when malaise, myalgias, and nausea developed, with five episodes of vomiting. The patient's parents took her to the emergency department of another hospital. On evaluation, the blood pressure was 125/85 mm Hg and the heart rate 120 beats per minute. The patient was actively vomiting and had dry mucous membranes. She appeared to be well developed and nourished. Laboratory test results are shown in Table 1. Intravenous fluids were administered. The symptoms abated slightly, and the patient was discharged home.

During the next 10 days, the vomiting continued, and intermittent fever, epigastric pain, nasal congestion, and cough developed. When the episodes of vomiting increased in frequency, the patient's parents took her back to the emergency department of the other hospital. The blood pressure was 162/112 mm Hg and the heart rate 98 beats per minute. The patient reported nausea; mild tenderness was noted in the left upper quadrant of the abdomen. The blood level of potassium was 2.7 mmol per liter (reference range, 3.5 to 5.1). Tests for severe acute respiratory syndrome coronavirus 2, influenza A and B viruses, and respiratory syncytial virus were negative. Other laboratory test results are shown in Table 1. Intravenous ondansetron, fluids, and potassium were administered, and the patient was transferred to this hospital for further treatment.

On arrival at this hospital, the patient reported a 3-month history of intermittent headaches and dizziness that had progressively increased in frequency from once or twice weekly to daily. For years, she had had intermittent chest pain, palpitations, and tremors that were associated with anxiety. These symptoms had increased in severity and frequency in the 10 days before the current admission. The patient had had a decreased appetite that was associated with depression, with an unintentional weight loss of 7 kg in the year before this admission. She reported no diarrhea, constipation, joint pain, or cold or heat intolerance. There was

Table 1. Laboratory Data.*

Variable	Reference Range, Other Hospital	10 Days before Current Admission, Other Hospital	Day of Current Admission, Other Hospital	Reference Range, This Hospital†	On Current Admission, This Hospital
Blood					
Hematocrit (%)	33.9–45.0	41.3	38.3	36.0–46.0	35.6
Hemoglobin (g/dl)	11.3–15.0	14.2	13.3	12.0–16.0	12.8
Platelet count (per μl)	189,000–400,000	606,000	559,000	150,000–450,000	458,000
White-cell count (per μl)	4800–10,800	11,000	9500	4500–13,500	7650
Differential count (per μl)					
Neutrophils	1900–8200	9400	5500	1800–8000	3890
Lymphocytes	300–4200	1100	2900	1500–6500	2890
Monocytes	200–800	400	800	200–1500	670
Sodium (mmol/liter)	136–145	134	132	135–145	132
Potassium (mmol/liter)	3.5–5.1	3.1	2.7	3.4–5.0	2.9
Chloride (mmol/liter)	98–107	89	86	98–108	91
Carbon dioxide (mmol/liter)	21–32	30	34	23–32	30
Urea nitrogen (mg/dl)	7–18	9	8	8–25	7
Creatinine (mg/dl)	0.7–1.1	0.7	0.6	0.60–1.50	0.58
Glucose (mg/dl)	74–160	123	117	70–110	98
Calcium (mg/dl)	8.5–10.5	9.7	9.1	8.5–10.5	8.3
Aspartate aminotransferase (U/liter)	12–45	—	21	—	—
Alanine aminotransferase (U/liter)	8–34	—	9	—	—
Alkaline phosphatase (U/liter)	45–504	—	147	—	—
Total bilirubin (mg/dl)	0.2–1.0	—	0.7	—	—
Direct bilirubin (mg/dl)	0.0–0.2	—	<0.2	—	—
Albumin (g/dl)	3.4–5.2	—	3.6	—	—
Total protein (g/dl)	6.4–8.2	—	7.4	—	—
C-reactive protein (mg/liter)	—	—	—	0.0–8.0	57.6
Erythrocyte sedimentation rate (mm/hr)	—	—	—	0–19	98
Urine					
Color	—	—	—	Yellow	Colorless
Clarity	—	—	—	Clear	Clear
Blood	—	—	—	Negative	3+
Glucose	—	—	—	Negative	Negative
Bilirubin	—	—	—	Negative	Negative
Protein	—	—	—	Negative	3+
Ketone	—	—	—	Negative	Negative
Nitrite	—	—	—	Negative	Negative
Leukocyte esterase	—	—	—	Negative	Negative
Specific gravity	—	—	—	1.001–1.035	1.008
Urobilinogen	—	—	—	Negative	Negative
White cells (per high-power field)	—	—	—	0–9	<10
Red cells (per high-power field)	—	—	—	0–2	>100

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for children who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

no acne or hair loss, but the patient described excessive hair growth on the back. She had no rash, but she described skin-colored stretch marks under the breasts and on the buttocks. The menstrual cycles were regular; she was currently menstruating.

The patient's depression had led to a suicide attempt and psychiatric hospitalization 5 months before the current admission. She took no medications, had no known allergies, and had received all routine childhood vaccinations. The patient's weight had been in the 50th percentile at 8 years of age, had increased to the 75th percentile by 12 years of age, and then had decreased to the 50th percentile at 13 to 14 years of age. The patient was in high school and lived in a suburban area of New England with her parents, two siblings, and several members of her extended family. She reported that she was not sexually active. She vaped flavored nicotine but did not smoke cigarettes, drink alcohol, or take illicit drugs. Her parents and siblings were healthy.

On examination, the temporal temperature was 36.9°C, the blood pressure 180/116 mm Hg, the heart rate 103 beats per minute, and the respiratory rate 20 breaths per minute. The patient appeared healthy and comfortable, and she was alert, awake, oriented, and cooperative. The neck was supple; a bruit was noted over the left carotid artery. Examination of the heart and lungs showed no abnormalities. The abdomen was nondistended, nontender, and soft, without organomegaly or masses. No acne or excessive hair was present on the face. No rash was noted on exposed skin, and there was no pain or swelling in the joints. The legs were warm, without swelling. The blood level of potassium was 2.9 mmol per liter (reference range, 3.4 to 5.0), the white-cell count 7650 per microliter (reference range, 4500 to 13,500), and the erythrocyte sedimentation rate 98 mm per hour (reference range, 0 to 19). Other laboratory test results are shown in Table 1. Intravenous hydralazine and oral potassium chloride were administered, and treatment with amiloride was started.

On the second hospital day, the headache and vomiting resolved, but the blood pressure remained elevated. Intravenous hydralazine was administered intermittently, amiloride therapy was continued, and amlodipine therapy was started. That evening, the patient reported blurry vision. While she was speaking to a health care worker, she abruptly stopped talking and stared. The blood pressure had increased to 182/111 mm Hg, and the patient was not responsive to voice or sternal rub. The pupils were round and reactive to light. After approximately 15 minutes, she became alert and oriented, but the blurry vision persisted. She was transferred to the pediatric intensive care unit for further treatment.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Rachel R. Osborn: This 14-year-old adolescent girl with a history of depression presents with either acute or intermittent hypertension in the context of hypokalemia and elevated inflammatory markers, including the erythrocyte sedimentation rate and C-reactive protein level. She also has subacute weight loss, and her current presentation overlays several months of chronic headaches, as well as years of intermittent chest pain and palpitations. Although the combination of headache, vomiting, and blurry vision prompts consideration of increased intracranial pressure, her hypertension alone is severe enough to explain many of her presenting symptoms, and increased intracranial pressure would not explain hypokalemia.

Given the acute neurologic changes that were noted on the second hospital day, when the patient abruptly stopped talking and stared, the presentation could be characterized as a hypertensive emergency, which warrants a thorough investigation for causes of secondary hypertension. I will focus on causes of secondary hypertension in the adolescent population, recognizing that the potential diagnosis should also account for the presence of electrolyte abnormalities and elevated inflammatory markers.[1]

GLOMERULONEPHRITIS

Hypertension is common in patients with glomerulonephritis, and the presence of hematuria and proteinuria in this patient is suggestive of glomerular disease.[2] IgA vasculitis (also known as Henoch–Schönlein purpura), poststreptococcal glomerulonephritis, and lupus nephritis are the most common causes of glomerular disease in children.

IgA vasculitis can result in abdominal pain and vomiting. However, the characteristic rash (palpable purpura) is absent in this patient, and IgA vasculitis is less common in adolescents than in younger children.

Children with poststreptococcal glomerulonephritis have a recognized prodromal illness (pharyngitis or impetigo) in 75% of cases, and such illness was not observed in this patient.[3] In addition, poststreptococcal glomerulonephritis is typically self-resolving, whereas this patient has a progressively worsening disease course.

A diagnosis of systemic lupus erythematosus could explain the patient's symptoms, history of neuropsychiatric disease, and elevated inflammatory markers. However, several aspects of her presentation argue against glomerulonephritis as the cause of her hypertension.

In patients with glomerulonephritis, hypertension is typically related to volume retention. This patient has a normal serum creatinine level and has no signs of fluid retention. The fact that she is currently menstruating offers an alternative explanation for the blood in her urine. Furthermore, the presence of hypokalemia and metabolic alkalosis suggest that the patient's hypertension is most likely mediated by aldosterone. Abnormalities in the renin–angiotensin–aldosterone system rarely contribute to hypertension in patients with glomerulonephritis.

NEUROENDOCRINE DISEASE

Cushing's syndrome (hypercortisolism) can cause hypertension in children, and it is frequently associated with psychiatric disease, which was seen in this patient. However, Cushing's syndrome typically promotes hair growth in a hirsute pattern, and this patient reported hair growth only on the back. In addition, Cushing's syndrome typically results in violaceous striae, and this patient reported skin-colored stretch marks. Cushing's syndrome is more frequently associated with weight gain than with weight loss; this patient had recently lost weight. It is notable that primary adrenal tumors causing Cushing's syndrome can have both cortisol- and aldosterone-secreting cells, but overall, Cushing's syndrome is unlikely in this patient.

The patient's history of hypertension, palpitations, headaches, and elevated inflammatory markers is consistent with a diagnosis of pheochromocytoma. Pheochromocytoma also could explain the seemingly intermittent nature of her hypertensive episodes. However, in patients with pheochromocytoma, hypertension is caused by adrenaline precursors from neuroendocrine cells outside the adrenal glands, which do not affect aldosterone activity. Thus, pheochromocytoma would not explain this patient's hypokalemia or metabolic alkalosis.

Primary aldosteronism occurring with adrenal adenoma or with the adrenal gland hyperplasia commonly seen in patients with idiopathic hyperaldosteronism can cause hypertension, hypokalemia, and metabolic alkalosis.[4,5] However, patients with primary aldosteronism typically have more marked hypokalemia and less severe hypertension than those seen in this patient. Furthermore, primary aldosteronism is not associated with elevated inflammatory markers.

VASCULAR DISEASE

Secondary aldosteronism due to abnormalities in the renal vasculature is the most likely cause of this patient's presentation with severe hypertension, mild hypokalemia, and metabolic alkalosis. The presence of hypertension and elevated inflammatory markers in a patient with constitutional symptoms should always prompt consideration of systemic sclerosis, which can lead to a life-threatening hypertensive emergency and kidney failure. However, this patient does not have the characteristic skin findings of systemic sclerosis. Furthermore, the scleroderma renal crisis that manifests in patients with systemic sclerosis involves the small renal arteries; the finding of a carotid bruit in this patient suggests that she has large-vessel disease. Behçet's disease is an inflammatory vasculitis that can affect the large vessels, but hallmark findings such as oral and genital ulcers and uveitis are absent in this patient.

Fibromuscular dysplasia is a common cause of renal artery stenosis with secondary aldosteronism. It is associated with severe hypertension, given that both aldosterone and angiotensin II are active and affect the process of vasoconstriction. Fibromuscular dysplasia can involve other vessels, which could explain this patient's carotid bruit. However, it is not an inflammatory process, and elevated inflammatory markers would not be expected. In addition, fibromuscular dysplasia would not explain the history of fever, malaise, and unintentional weight loss that preceded the patient's current presentation.

In this patient's case, the problem representation can be summarized as an adolescent girl presenting with a hypertensive emergency in the context of a probable inflammatory large-vessel vasculitis and secondary aldosteronism. This problem representation prompts consideration of Takayasu arteritis, a rare diagnosis with an illness script that has considerable overlap with that of giant-cell arteritis. Children with Takayasu arteritis present with hypertension in 83% of cases, and neurologic symptoms such as blurry vision, headache, and dizziness are common.[6] The prodromal phase of this inflammatory vascular disease is often characterized by nonspecific symptoms. The aortic arch is often involved, which can lead to discordant blood pressures between the two arms. This patient had a normal blood pressure in the emergency department of the other hospital but has severe hypertension at this hospital — a discrepancy that may be explained by measurement of the blood pressure in different arms.

On the basis of this patient's clinical findings, I suspect that she has Takayasu arteritis. To establish the diagnosis, I would obtain angiographic images to look for vascular stenosis, occlusion, or aneurysm formation.

DR. RACHEL R. OSBORN'S DIAGNOSIS

Takayasu arteritis.

HOSPITAL COURSE

Dr. Weizhen Tan: On renal ultrasound images, which had been obtained on the first hospital day, the right and left kidneys measured 11.8 cm and 9.9 cm, respectively — a size discrepancy of greater than 1 cm. This asymmetry was consistent with a subacute process and suggested renal artery stenosis, which can adversely affect the renin–angiotensin–aldosterone system. Dysfunction of this system could explain many of the findings in this patient.

In patients with renal artery stenosis, the renin and aldosterone levels are both elevated, and the aldosterone-to-renin ratio is therefore expected to be low.[7] This patient had elevated blood levels of renin (57.0 ng per milliliter per hour; reference value, <2.1) and aldosterone (89 ng per deciliter; reference value, <21), with an aldosterone-to-renin ratio of 1.56 (reference value, >20). Computed tomographic angiography (CTA) was performed to evaluate for renal artery stenosis.

DIAGNOSTIC IMAGING

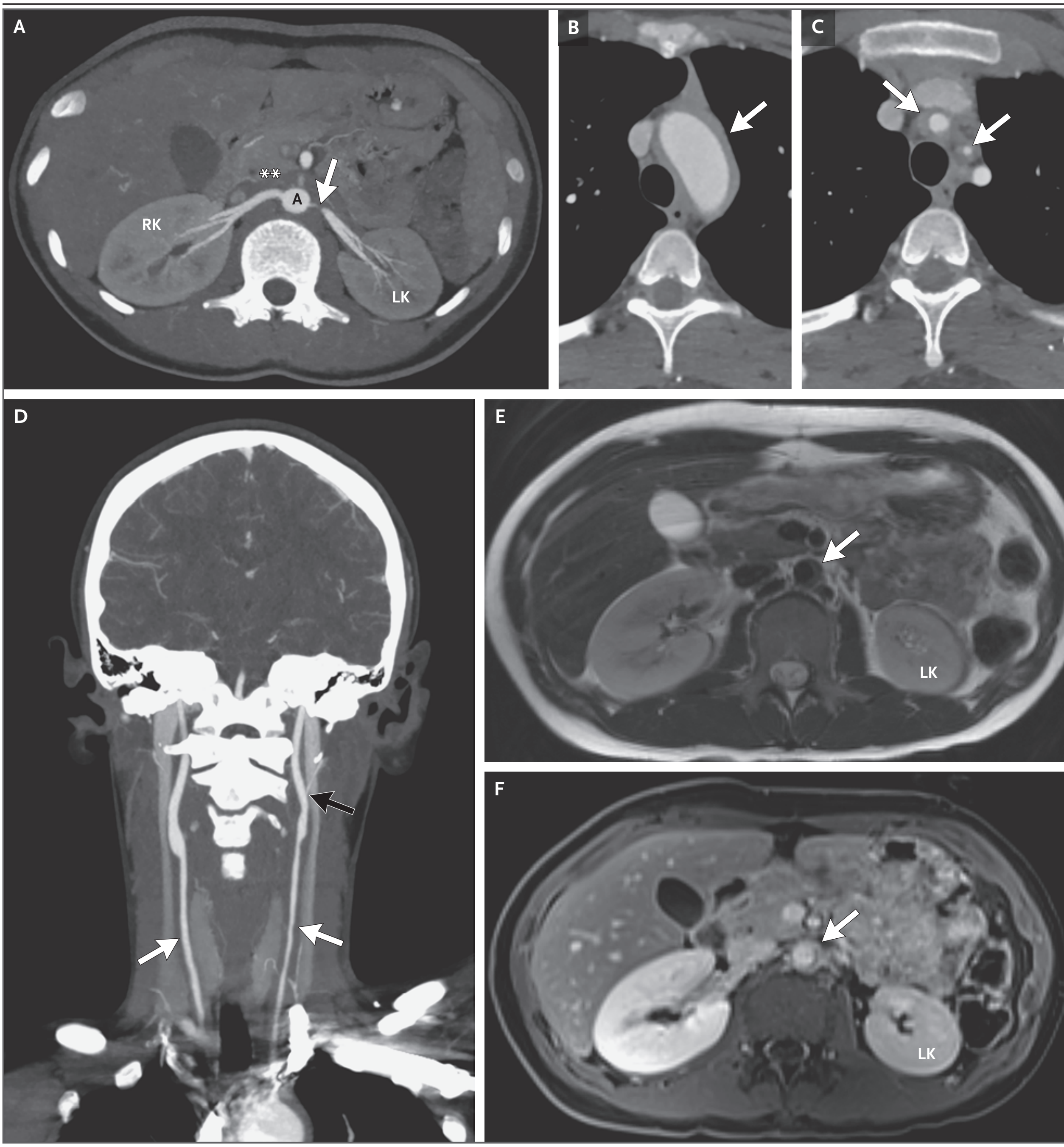
Dr. Evan J. Zucker: CTA of the abdomen, performed after the administration of intravenous contrast material, revealed severe stenosis of the left renal artery, with hypoenhancement and atrophy of the left kidney that probably resulted from chronic vascular obstruction (Fig. 1A). Renal artery stenosis could explain the patient's hypertension. However, prominent soft-tissue thickening and stranding was also present, with slight narrowing of the abdominal aorta. These additional features of vascular inflammation strongly suggested an underlying large-vessel vasculitis. In this adolescent girl, Takayasu arteritis was the most likely possibility.

To investigate for other potential sites of involvement, contrast-enhanced CTA of the head, neck, and chest was subsequently performed. Multifocal arterial wall thickening and luminal narrowing involving the aortic arch, great vessel origins, and carotid arteries were observed (Fig. 1B, 1C, and 1D). These findings are characteristic of Takayasu arteritis. Contrast-enhanced magnetic resonance angiography (MRA) of the head,

Figure 1 (facing page). Imaging Studies.
Contrast-enhanced computed tomographic angiography (CTA) of the abdomen was performed. An axial oblique maximum-intensity-projection (MIP)–reformatted image (Panel A) shows severe narrowing of the left renal artery (arrow). Abnormal periaortic soft-tissue thickening is present (asterisks), with a slightly decreased luminal caliber of the abdominal aorta (A). The left kidney (LK) has notable hypoenhancement and atrophy as compared with the right kidney (RK), a finding consistent with chronic, high-grade renovascular obstruction on the left side. Contrast-enhanced CTA of the head, neck, and chest was subsequently performed. Serial axial images (Panels B and C) show abnormal wall thickening (arrows) of the thoracic aortic arch and the origins of the brachiocephalic and left common carotid arteries. In addition, severe luminal narrowing of the left common carotid artery is present. A coronal oblique MIP-reformatted image (Panel D) shows long-segment wall thickening with associated luminal narrowing (white arrows) throughout both common carotid arteries; the finding is more prominent on the left side than on the right side. Severe stenosis is present at the origin of the left external carotid artery (black arrow). Contrast-enhanced magnetic resonance angiography (MRA) of the abdomen was also performed. An axial T2-weighted double inversion recovery image (Panel E) and a nephrographic phase, postcontrast image (Panel F) show abnormal abdominal aortic wall thickening (arrows) with associated contrast enhancement, a finding suggestive of active inflammation. Hypoenhancement and atrophy of the left kidney is visible. Findings on MRA of the head, neck, and chest (not shown) were concordant with previous CTA findings.

neck, chest, and abdomen was also performed to help better characterize disease activity. The MRA findings were concordant with the previous CTA findings. MRA was also notable for enhancement within areas of vessel-wall thickening, a finding supportive of active vasculitis (Fig. 1E and 1F). Imaging of the head was normal.

Imaging is the preferred diagnostic method to establish the hallmark findings of Takayasu arteritis, including vascular wall thickening and enhancement, stenosis, and aneurysm formation in a characteristic distribution, involving the thoracoabdominal aorta or its major branches.[8-10] Although the brain is not always affected and this patient had no evidence of parenchymal involvement, an intracranial vascular evaluation was nonetheless prudent owing to her neurologic symptoms.[11]



Among the options for diagnostic imaging for Takayasu arteritis, CTA or MRA is typically chosen. When feasible, MRA offers additional characterization of the vessel wall that correlates with disease activity.[10] Unlike CTA, MRA does not involve exposure to ionizing radiation and thus is favored in children. However, CTA can usually be performed more expediently and has highly reproducible results, whereas MRA is often more technically challenging, time consuming, capacity-constrained, and costly. Conventional angiography is possible, but this approach is invasive and cannot provide vessel-wall delineation.[8] ^{18}F-fluorodeoxyglucose positron-emission tomography (PET) may show radiotracer uptake in the vessel wall that is associated with active inflammation, but the added prognostic benefit remains unclear. Clinical research is ongoing regarding the use of hybrid PET–magnetic resonance imaging as a promising option for diagnostic imaging for pediatric Takayasu arteritis.[10]

IMAGING DIAGNOSIS

Takayasu arteritis.

DISCUSSION OF MANAGEMENT

Dr. Tan: The initial management of acute renal artery stenosis is focused on preventing further symptoms of hypertensive emergency and end-organ damage. The goal is to reduce the blood pressure while minimizing the risk of hypoperfusion. For acute hypertensive emergency in children, we aim to establish a blood pressure in the 95th percentile (using a target based on height, age, and sex or a target of 130/80 to 140/90 mm Hg for children ≥13 years of age), with 25% of the planned reduction in systolic and diastolic pressure occurring within 8 hours and the remainder occurring within the subsequent 12 to 48 hours.[1] We avoid the use of angiotensin-converting–enzyme inhibitors and angiotensin-receptor blockers in patients with renal artery stenosis. In this patient, we administered a continuous infusion of intravenous nicardipine, which easily facilitated adjustment of her blood pressure over the course of several days.

After the initial phase, the patient's treatment was slowly transitioned to an oral calcium-channel blocker. A beta-blocker, a potassium-sparing diuretic agent, and supplemental potassium were added. Long-term management of renal artery stenosis involves treatment of the underlying disease, which in this case was Takayasu arteritis.

Dr. Jonathan S. Hausmann: Takayasu arteritis is a rare chronic large-vessel vasculitis that affects the aorta and its major branches, leading to clinically significant complications from vascular stenosis, occlusion, and aneurysm formation. In children, Takayasu arteritis has an incidence of 0.4 to 6.0 in 1 million, with a mean age at onset of 12 years, although the disease has even been described in infancy.[12] Cases develop in girls and boys at a ratio of 2.5:1 to 3:1.[13]

In this patient, weight loss and other symptoms occurred for months before the diagnosis was established. Diagnostic delays are common in children with Takayasu arteritis because the condition is extremely rare and the presentation is often vague. Children may present with vascular manifestations such as hypertension, bruits, and a reduced pulse. However, they may also present with nonspecific symptoms such as fever, malaise, weight loss, abdominal pain, and headache, which are often associated with systemic inflammation.[13,14]

Takayasu arteritis in children often involves the abdominal aorta and renal arteries, as seen in this patient; Takayasu arteritis in adults more commonly affects the branches of the aortic arch. These anatomical differences may explain why children with Takayasu arteritis may present with hypertension, elevated creatinine levels, and abdominal pain, whereas adults with Takayasu arteritis more often have loss of pulse and limb claudication.[15]

The American College of Rheumatology and the European Alliance of Associations for Rheumatology developed diagnostic criteria to distinguish Takayasu arteritis from other vasculitides in adults (Table 2).[9] Although the classification criteria have not been clinically validated for use in children, several of the criteria were met in this patient, including the absolute requirements of being 60 years of age or younger and having evidence of vasculitis on imaging, as well as additional clinical and imaging criteria: female sex, vascular bruits, involvement of three or more arterial territories, symmetric involvement of paired arteries, and abdominal aortic involvement with renal vessel involvement.

The primary goals of treatment in patients with Takayasu arteritis are to control inflammation, maintain vascular patency, and manage hypertension effectively. Treatment recommendations have been published for children and adults,[16] although most of the recommendations are conditional owing to the rarity of the disease and the absence of high-quality data from randomized clinical trials. For patients with active, severe disease, the use of high-dose oral glucocorticoids is recommended. Because glucocorticoid monotherapy often leads to relapse and clinically significant long-term side effects, especially in children, the use of a glucocorticoid-sparing, disease-modifying agent is also recommended. Guidelines do not state a preferred immunosuppressive agent; methotrexate, azathioprine, or a tumor necrosis factor inhibitor can be used. However, results from small, retrospective studies involving children suggest that the 2-year flare-free survival seen with biologic therapies (80%) is twice that seen with nonbiologic therapies.[17]

For the management of vascular complications of Takayasu arteritis, medical interventions are generally recommended, rather than surgical or catheter-based interventions. However, in patients who have renovascular hypertension that is refractory to medical management or is causing worsening kidney function, vascular interventions may be warranted. If surgery is pursued, the procedure should be performed when the disease is quiescent. Although children with Takayasu arteritis frequently undergo endovascular interventions for the treatment of renovascular hypertension, half these patients have restenosis within 1 year.[13]

Table 2. Classification Criteria for Takayasu Arteritis in Adults.*

Criterion	Status or Points	Status or Points in This Patient
Absolute requirements		
Age ≤60 yr at time of diagnosis	Yes	Yes
Evidence of vasculitis on imaging	Yes	Yes
Additional clinical criteria		
Female sex	+1	+1
Angina or ischemic cardiac pain	+2	—
Arm or leg claudication	+2	—
Vascular bruit	+2	+2
Reduced pulse in arm	+2	—
Carotid-artery abnormality	+2	+2
Difference in systolic blood pressure of ≥20 mm Hg between arms	+1	—
Additional imaging criteria		
No. of affected arterial territories		
One	+1	—
Two	+2	—
Three or more	+3	+3
Symmetric involvement of paired arteries	+1	+1
Involvement of the abdominal aorta with involvement of a renal or mesenteric vessel	+3	+3

* Adapted from Grayson et al.[9] When a diagnosis of medium-vessel or large-vessel vasculitis has been made, these classification criteria can be applied to determine whether the patient has Takayasu arteritis. Diagnoses that mimic vasculitis should be ruled out before the criteria are applied. Adults who meet the absolute requirements and have a score of 5 or higher are classified as having Takayasu arteritis. The classification criteria have not been clinically validated for use in children.

FOLLOW-UP

Dr. Tan: In coordination with the rheumatology team, the patient received glucocorticoid therapy, along with methotrexate and an infusion of infliximab. After 5 months of treatment, repeat renal ultrasonography showed atrophy of the left kidney. Repeat MRA showed poor growth of the left kidney and poor perfusion, with persistent critical stenosis of the left renal artery. The interventional radiology team was consulted, and after a multidisciplinary discussion, a decision was made to pursue angioplasty, which was performed 6 months after the initial presentation.

Dr. Zucker: Digital subtraction angiography confirmed severe stenosis of the left renal artery, which abated substantially after balloon angioplasty (see video, available with the full text of this article at NEJM.org).

A video showing angiography before and after treatment is available at NEJM.org

Dr. Tan: After angioplasty, hypertension resolved, and the patient was discharged while receiving no antihypertensive medications. At 24 months after the initial presentation, her blood pressure remained normal. Repeat Doppler ultrasonography showed continued stenosis of the left renal artery. Treatment with supplemental potassium was indicated for ongoing mild hypokalemia. During this time, the patient had become sexually active. Long-acting reversible contraception was initiated, but the patient discontinued contraception owing to personal preference. She continues to be counseled regarding the benefits of contraception with the use of teratogenic medications.

FINAL DIAGNOSIS

Takayasu arteritis.

This case was presented at the postgraduate course, “New England Pediatric Hospital Medicine 2024,” directed by Chadi El Saleeby and Kerstin Zanger.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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